BASIM HUSSAIN

AI Engineer | Software Engineer

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Transferable Iqama | Available immediately | English (fluent), Arabic (professional working), Urdu and Hindi (native)

PROFILE

AI Engineer with a full-stack background, building and shipping AI features inside real customer-facing apps. Most recent work covers LLM chat and voice agents, RAG pipelines, a fine-tuned Arabic language model and automation that removed repetitive work for sales and support teams. Earlier, I built a consumer booking app used by around 5,000 people and the support chatbot that sat behind it. Comfortable across Python and JavaScript, from model evaluation and API design through to deployment, and used to working closely with product and engineering teams on bilingual Arabic and English products for the Saudi market.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, PHP, Java, SQL

AI / ML: LLM application development, RAG pipelines (LangChain, LlamaIndex, pgvector), fine-tuning (QLoRA, LoRA, PEFT), AI agents and function calling, prompt engineering, model evaluation and observability (RAGAS, Langfuse), Arabic NLP, OpenAI API, HuggingFace, PyTorch, FastAPI

Backend and APIs: Node.js, Express, Fastify, Laravel, Spring Boot, REST, WebSockets, Server-Sent Events

Web and Mobile: React 19, React Native (Expo), Vue.js, Nuxt.js, Tailwind CSS, Vuetify, i18n and RTL layouts

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, pgvector, Firebase, Prisma

Infrastructure: Docker, GitHub Actions CI/CD, AWS (S3, Lambda, CloudWatch), Git, Sentry, n8n

Enterprise: Microsoft Power Platform (Power Apps, Power Automate, Dataverse, Power BI)

PROFESSIONAL EXPERIENCE

AI Software Engineer

Basmat Alomran Co. | Riyadh, Saudi Arabia | Mar 2026 – Present

- Built and deployed voice and chat AI agents that handle inbound enquiries and lead qualification, with call summaries written back to the CRM automatically. Around 70% of repetitive first-contact enquiries no longer reach the sales team.
- Developed an LLM pipeline that turns free-text customer requirements into structured, service-based quotations, and an n8n WhatsApp automation that brought routine response times down from roughly 4 hours to under 30 seconds.
- Shipped bocsmart.com, a bilingual Arabic and English corporate platform with a live Smart AMC Estimator that returns a quote in under 2 seconds.
- Designed and built a company-wide ERP from scratch covering HR, finance, procurement, projects and field operations, with role-based access, multi-stage approvals and real-time dashboards that replaced manual daily reports across 3 active sites.

Stack: *Vue.js, Node.js, Express, MySQL, Docker, CI/CD, LLM and Voice APIs, n8n*

Founding AI Engineer (Full-Stack)

Agent Maven | Riyadh, Saudi Arabia | Oct 2025 – Mar 2026

- Architected a Voice AI SaaS platform that orchestrates conversational agents (Vapi) on a Node.js and PostgreSQL backend designed for 100K+ calls a day.
- Set up a RAGAS evaluation loop (faithfulness 0.93, answer relevancy 0.90) with automated regression tests on every knowledge-base update, so hallucination regressions were caught before they reached production.
- Built a real-time React 19 dashboard on Socket.io for live call streaming, sentiment visualisation and performance metrics.
- Containerised every service with Docker and built a multi-stage GitHub Actions pipeline with test gates. No failed production deployments through launch.

Stack: *Node.js, Express, PostgreSQL, Supabase, React 19, Redux, Socket.io, Vapi AI, Tailwind, Docker*

Full-Stack Engineer (three-month MVP engagement)

Tamkeen Middle East | Riyadh, Saudi Arabia | Jul 2025 – Oct 2025

- Led end-to-end delivery of Athar, a bilingual platform connecting Saudi retail investors to Vision 2030-aligned social impact funding, working directly with the product owner and design team to scope, prioritise and ship the MVP on schedule.
- Built Saudi-compliant KYC onboarding with multi-stage verification, SMS OTP, an AWS S3 document pipeline and a Laravel Nova admin console.
- Engineered the Laravel 10 REST API with Sanctum authentication and rate limiting, and the Vue.js and Nuxt.js frontend with full i18n and RTL support, with no layout breakage in either language.

Stack: *Laravel 10, Vue.js, Nuxt.js, Vuetify, MySQL, AWS S3, Laravel Sanctum*

Software Engineer

Reshantech Sports Pvt. Ltd. | India (remote) | Jul 2024 – Jul 2025

- Shipped a greenfield React Native (Expo, TypeScript) turf-booking app with Firebase Auth and Sentry monitoring, reaching around 5,000 active users with over 99% crash-free sessions.
- Built an AI chatbot for customer support and booking assistance covering FAQs, slot availability and booking guidance, which took roughly 60% of support queries off human agents.
- Engineered the Node.js booking engine with optimistic locking. Zero double-bookings across 10,000+ transactions under concurrent load.

Stack: *React Native, TypeScript, Node.js, Express, MongoDB, Firebase, Sentry*

SELECTED PROJECTS

Arabic Sentiment Fine-Tune

Llama 3 8B, QLoRA, HuggingFace, FastAPI

- Fine-tuned Llama 3 8B with QLoRA (4-bit NF4) on 12K Arabic tweets, lifting 3-class sentiment F1 from 0.58 to 0.87 across 14 tracked training runs. Deployed the quantised model behind a FastAPI endpoint on HuggingFace Spaces with latency and memory monitoring.

ScentWell, AI Recommendation Engine

React Native, Node.js, pgvector, OpenAI, Supabase

- Built a RAG recommendation engine over pgvector with a GPT-4o conversational agent grounded by live web search and Reddit data. Instrumented with Langfuse tracing and RAGAS evaluation (faithfulness 0.90, relevancy 0.88), cutting hallucinations by 35%.

Medical Image Translation using GANs

PyTorch, Computer Vision, 2nd Prize, AMU ZHCET Project Exhibition 2025

- Designed a pix2pix-style conditional GAN for MRI-to-CT translation aimed at removing redundant scans in clinical workflows, validated with SSIM. Selected from all final-year projects for the award.

Also: Saudi Business Research Agent (autonomous 5-tool agent with PostgreSQL memory, full company research in under 16 seconds), Mise (multi-tenant restaurant operations platform on Fastify, Postgres and Redis), Microsoft Power Platform suite. Full detail at github.com/Basimhussain.

EDUCATION AND RESEARCH

B.Tech. Computer Engineering

Aligarh Muslim University | May 2021 – May 2025

- Research Experience for Undergraduates (REU), Indo-US collaboration jointly awarded by The Ohio State University and AMU, 2024. Project: AI Gesture Mouse for macOS (OpenCV, MediaPipe).

CERTIFICATIONS AND AWARDS

- AWS Certified Cloud Practitioner (CLF-C02), 2026
- REU Certificate of Achievement, The Ohio State University and Aligarh Muslim University, 2024
- 2nd Prize, Project Exhibition (Medical GAN), ZHCET, Aligarh Muslim University, May 2025
- LinkedIn Learning: Prompt Engineering for Generative AI, Statistics Foundations, Data Analytics Career Skills, Excel Expert, 2025